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**Advanced Hemodynamic Modelling and Simulation of Flow
Diverters in the Treatment of Intracranial Aneurysms**

Soutenue par

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Dedication

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Chapter 1

Patient Study

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With the presented chapters on theory of fluid dynamics, the brain arterial system and the description of a new virtual stent deployment method, one complex patient study is going to be analyzed. The case has already been used in the stent deployment to assess the performance of the method. Here we will further develop on the hemodynamic effects in it and explore the implications of the FDS treatment, which, sadly, can be related to the cause of death of the patient. To support our and the clinicians understanding, we will evaluate pre- and poststented configurations in regards to common remodelling indicators and lastly consider a novel simulation including wall compliance. The latter simulation has been done in cooperation with Aurèle Goetz, whose work is dedicated to the Fluid-Structure Interaction (FSI) modelling of intracranial aneurysms.

1.1 Medical background

1.1.1 Patient medical record

The patient is a woman born in January 1963, with no history of smoking, hypertension, nor family history of intracranial aneurysms. Vertigo along with anxiety and a strange sensation lead her to seek consultation of a neurologist. After a first consultation, she was appointed to do an MRI examination of the crane, revealing a giant intracranial aneurysm compressing the left temporal lobe¹. Her neurologist referred her to a neuro-interventionalist in the same month for further evaluation. A cerebral arteriography confirmed the existence of a giant aneurysm in the supraclinoid segment of the left ICA. The englobing of the parent artery by the aneurysm cause a delayed opacification of the left middle cerebral artery. The small anterior communicating artery was visible during compression of the left ICA.

Based on the presence of symptoms and a menace of rupture, endovascular treatment was proposed to the patient, to which she agreed during the following medical consultation. One week before the procedure, she began taking aspirin (75 mg/day) and prasugrel² (10 mg/day) to reduce the risk of blood clots. The procedure started with a balloon test occlusion, where a balloon was temporarily inflated in the artery to block blood flow. A guiding catheter was placed in both the right and left internal carotid arteries. A Scepter XC 4–11 balloon was positioned in the left ICA upstream of the aneurysm and inflated to stop blood flow. Contrast injection via the right catheter showed a delay of over one second in the opacification of the left cerebral veins compared to the right, indicating that the left carotid artery could not be safely occluded. Therefore, preserving and reconstructing the left carotid artery with a flow-diverting stent was necessary. A 3 mm FRED Jr flow-diverter was de-

¹Temporal lobes: Regions of the brain situated on both halves above the ears.

²Prasugrel: Platelet aggregation inhibitor medication.

ployed from the left carotid terminus to the left ICA upstream of the aneurysm, immediately causing contrast stasis in the aneurysm sac. Upon waking up, the patient showed no neurological deficits and continued antiplatelet therapy. At her follow-up 1.5 months later, her clinical exam returned normal results. Six months after the procedure, a follow-up cerebral arteriography showed a patent³ stent with slight narrowing at its origin and unchanged opacification of the aneurysm sac. Continued dual antiplatelet therapy was advised, with a repeat arteriography scheduled for ten months post-procedure.

Eighth months after the intervention, the patient reported a sudden headache. A brain CT scan revealed spontaneous hyperdensity in the aneurysm, suggesting thrombosis (blood clot). A week later, she suffered a sudden coma, with a Glasgow Coma Scale score of 4, indicating severe brain injury. A CT scan revealed subarachnoid hemorrhage (bleeding in the space surrounding the brain), intraventricular hemorrhage, and a left temporal hematoma⁴. Cerebral arteriography showed persistent aneurysm opacification, a patent stent, and active contrast leakage from the aneurysm sac's postero-inferior part. A microcatheter was placed in the aneurysm sac through the stent mesh, and the sac was occluded with coils, halting the bleeding and completely occluding the aneurysm. Sadly, the patient passed away five days after this second intervention.

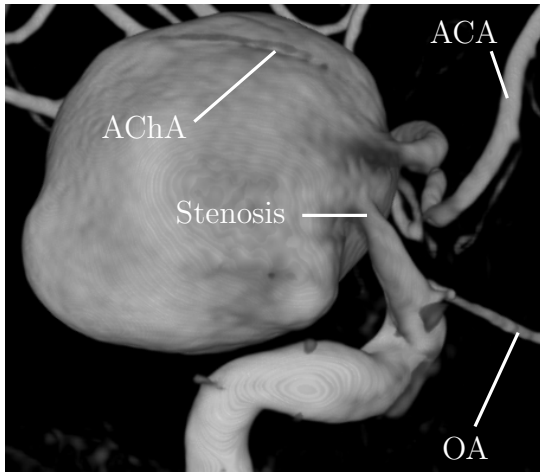


Figure 1.1: Annotated DSA image of the preoperative IA.

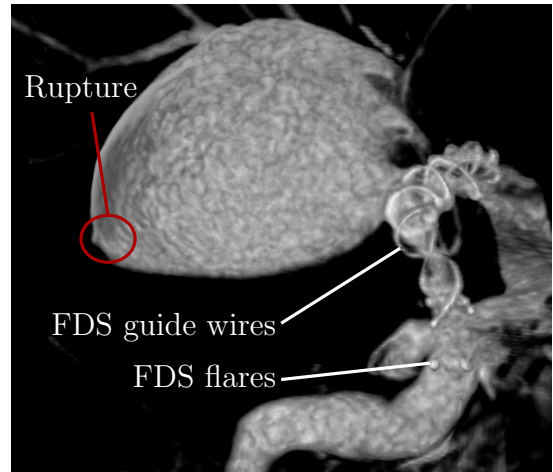


Figure 1.2: Annotated DSA image of the postoperative IA.

³Patent (stent): Maintaining the openness and integrity of the vessel or structure where it has been placed.

⁴Hematoma: A solid swelling of clotted blood within the tissues.

1.1.2 Artery topography

Circle of Willis

An MRA scan confirmed that the present CoW to be incomplete, missing the posterior communication ipsilateral to the aneurysm. The contralateral one, which appears to have a fetal nature, contributes most of the blood to the left posterior circulation. The anterior communication, although present, has a small, possibly hypoplastic size which was confirmed during a clamp test. The aneurysm is found at the C7 section of the ICA, 4mm before the terminal bifurcation of the vessel. The Anterior Choroidal Artery (AChA) emanates from it, however its flow is barely noticeable on DSA scans, as seen on the left of [Fig. 1.1](#).

Tampering

An important factor considering this case is the presence of Tampering (T) and vessel deformation around the aneurysm. The cavernous ICA (C3) has a diameter of 4.16 mm falling in line with common values [\[1\]](#). It experiences a continuous tampering over the communicating and ophtalmic segments leading to a narrow, nearly elliptical cross-section at the proximal aneurysm neck. This atypical cross-section translates into a 32% tampering or stenosis with respect to the C3 diameter (see [Table 1.1](#)). Past the aneurysm, the terminal section of the ICA narrows down to 2.15 mm, equivalently to a 48% stenosis with respect to the C3. This value clearly falls outside the standart deviation of the recorded cases by Baz *et al.* [\[1\]](#).

After deploying the stent, the vessel progressively changes its shape into a more regular, circular artery. The proximal section is narrower and fall below common values, as shown on [Table 1.1](#). The sections at the aneurysm have become nearly identical in diameter, attaining typically found sizes. Although it is not verifiable from the imaging, these effects are likely to be caused by the mechanical forces exerted by the stent and the progressive remodelling of the artery as well.

1.1.3 Aneurysm

Morphology

The aneurysm contained part of its parent vessel prior to the operation. It has a nearly round shape with a diameter of appoximately 25mm and a volume of 7170 mm³. In the course of the eight months after the intervention, the aneurysm underwent changes, similarly to the vasculature. The most notable changes are the size reduction on its lower and left sides, which rendered its shape heart-like. The FDS also reshaped the parent vessel, transforming it into a circular artery with the aneurysm neck being now clearly visible (instead of the vessel traversing the

	Preoperative		Postoperative		Literature[1]	
	d [mm]	T [%]	d [mm]	T [%]	d [mm]	T [%]
Cavernous ICA	4.16	-	3.77	-	4.20	-
Aneurysm proximal	2.84*	32	2.76	27	2.86	32
Aneurysm distal	2.15	48	2.71	28	2.70	36

Table 1.1: Vessel diameters and Tampering (T) of the vessels with respect to the cavernous ICA section (C3). Literature values were taken from a reference study considering values for women. The preoperative diameter of the vessel at the aneurysms entry () is estimated from the circle radius that is equivalent to its area. The preoperative distal vessel diameter falls outside the standart deviation of the cited work by Baz et al. [1].*

aneurysm). The next rotational angiographic images stem from the hours after rupture, showing a small blood leakage at the tip of the aneurysm, as shown in Fig. 1.2. The rupture site is adjacent to the third ventricle, which permits an extensive movement of the aneurysm at this site.

Movement

The displacement of the aneurysm wall induced by the cardiac pulse, could only clearly be seen after rupture. The whole aneurysm movement can be described similarly to an inflating and deflating balloon, with no noticeable displacement at the neck and a progressively bigger movement towards the tip. During systole, the rupture zone moved by an estimated 10% of the aneurysm length. The quantitative displacement was however difficult to measure due to the rotating video angiography (see the left of Fig. 1.3 for reference), so we propose the value rather as an order of magnitude.

During the segmentation work, it was seen that the post-rupture images were not as nitid as the preoperative ones, particularly due to the absence of saturated contrast agent in some vessels that are essential. This can be seen in Fig. 1.2, where the MCA is nearly hollow and only the marker wires nidicate the location and size of the vessel. Inspection of the videos taken during injection showed a remarkable response of the aneurysm to the rise in pressure caused by the ink bolus being flushed through the system. The latter is injected following medical protocol at a rate of 3 mL/s (60% of the mean flow rate of the ICA) during 7 s. At systole, ink penetrates the aneurysm without visible reflux of fluids leaving the bulge. Right after systole, the aneurysm deflates, flushing out blood that renders the MCA transparent (see the right of Fig. 1.3). Despite being an effect only seen during the injection and

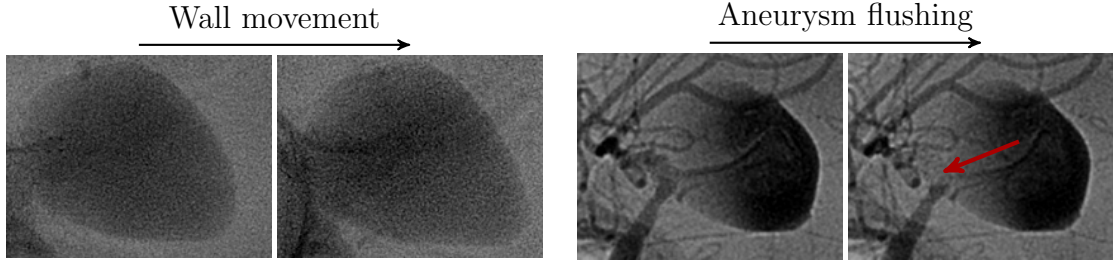


Figure 1.3: An attempt to display the different facets of the movement in the aneurysm. The frames of the contrast agent-filled aneurysm at diastole and systole (left) show a distension of the bottom and right faces of the aneurysm. The frames of the aneurysm during ink injection (right) at subsequent snapshots ($\Delta t = 0.17\text{ s}$) display the deflation of the aneurysm, flushing out notable quantities of ink-less blood and rendering the MCA transparent (red arrow).

therefore under increased pressure, this underscores the potential volume variation of the aneurysm and its pulsatility. When the ink injection is stopped, the reflux of the aneurysm is no longer visible, indicating a different type of fluid dynamics at play.

1.2 Simulation

The medical images have revealed important information about the case including the topology, morphology and some of the dynamics. These informations are now enriched by hemodynamics simulations, which will help to understand the pressure evolution and wall shear in the aneurysm. These two factors are potentially decisive in the remodelling and rupture dynamics as already discussed in previous publications on similar aneurysms[2, 3].

CFD setup

For the CFD study, there are two 3D angiography images available, taken before the stent deployment and after rupture. Both DSA images were segmented following the guidelines described in the previous chapter and meshed with a more refined mesh size ($h = 0.25\text{ mm}$). Due to the increased flow velocities at the stenosis, the preoperative case was run implicitly with a timestep of 0.5 ms . The convection-related instabilities at peak systole that motivate this strategy were not observed in the stented case, where semi-implicit time stepping was sufficient.

The inflow conditions were imposed through parabolic profiles according to averaged waveforms for people in the same age group as the patient [4]. We assumed the inflow

Case	h	Time integration	Δt	Comp. time
Preoperative CFD	0.25 mm	Implicit	0.0005 s	34 h
Postoperative CFD	0.25 mm	Semi-implicit	0.0005 s	18.5 h
Postoperative FSI	0.3 mm	Semi-implicit	0.001 s	50 h

Table 1.2: Computation times for the pre- and postoperative cases, including the FSI simulation from the PhD work of A. Goetz. Note that the Rigid-FSI comparisons that have been included from the cited work, stem from the same (coarser) fluid mesh.

conditions to remain the same pre- and postoperatively [2], but we acknowledge this to be a potential source of bias. Outlets were modelled using resistance models fit as described in Chapter 3. The targeted flow was chosen to depend on the cube of the MCA and ACA diameters, meeting Murray’s Law. As discussed in the before, the vessel walls in this case showed clear pulsatility on the CT scans, which possibly impacts the internal hemodynamics as well.

FSI setup

1.3 Results

The simulation jobs have been executed on 64 cores of the previously mentioned computing facility (see Chapter 3). The computation times and with the different integration modalities are shown in Table 1.2. The preoperative case caused a longer computation time due to the 2–5 subiterations per time step required for convergence. The postoperative case computed as a CFD simulation was faster, at approximately half the time per cardiac cycle. The FSI simulation of the ruptured aneurysm took with 50h per cardiac cycle the longest time, despite its lower spatial and temporal resolutions. The explicit coupling between the solid and fluid solvers, together with slower convergence rates for the solid solver, are the main reasons for this increase in computation time.

1.3.1 Pre-operative hemodynamics

The aneurysm hemodynamics before the insertion of the FDS are characterized by a high velocity jet (1.8 m/s) at the stenosis entering and rolling through the formation around a horizontal axis. It then spreads out over the aneurysm surface, generating a large vortex in the bulge. As a consequence, high OSI values form on the “eyes” of the vortex, as seen on Fig. 1.4. The WSS in these regions displays persistently an expansive properties with high divergences of the normalized WSS vectorfield.

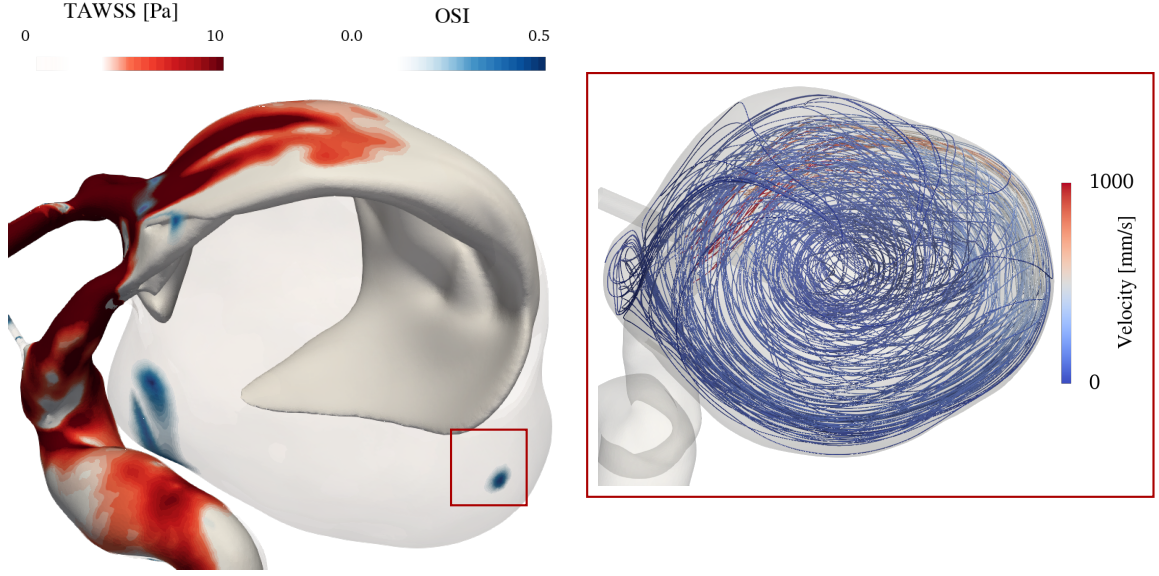


Figure 1.4: Preoperative hemodynamics of the aneurysm.

A pouch at the aneurysm's inner side stands out with high OSI and small TAWSS values, as a consequence of its irregular perfusion.

1.3.2 Post-operative hemodynamics

In the postoperative state, the FDS induces remodelling of the parent artery, which is now circular and does no longer traverse the aneurysm sack. Consequently, the impinging jet at systole is slowed down to half of its preoperative value (0.9 m/s). The volumetric flow traversing the stent at this same instant is 38% of the ICA's flowrate. This value however is difficult to compare to the preoperative state, since there the aneurysm is part of the vessel and a delimitation is difficult to find. For the same reason, it is difficult to quantify the bulge velocity reduction with respect to the preoperative case. The pulsatility index in the postoperative state was computed at 1.24, resembling common literature values for giant ICA aneurysms [5]. The PI belongs to the rather upper spectrum of values found, which is tied to two main factors. On one side, the impingement angle of the flow and size of the IA permit the formation of large vortices that retain a large proportion of their momentum throughout the cycle. The mean bulge-averaged velocity $\bar{u}$ therefore remains high, and so does also its minimum value. The second factor is the chosen patient waveform. The averaged curve presents two closely followed peaks at systole, instead of one more pronounced one. This leads to a more regular flow pattern and thus a lower PI value.

1.3.3 Pressure changes

Considered separately due to its alledged importance, is the alteration of the bulge pressures between the preoperative and postoperative (ruptured) cases. To that end, we have extracted the pressure along the centerline of the ICA and its branches, the ACA and MCA in both scenarios. Figure 1.5 shows on the upper row the pressures on the preoperative state, while the lower one does for the postoperative state. Within each row, the curves are identical up until the location of the aneurysm, since that portion of the centerline is shared between the centerlines computed for MCA and ACA. The different colorcoded curves mark the pressure progression at different flow rates of the cardiac cycle. The pressures have been offset to have a zero-pressure at diastole in the distal section of the MCAs.

Under the assumption of an unchanged inflow rate, the pressure drop across the system drops significantly between both simulations. In the case of the MCA at systole, for instance,

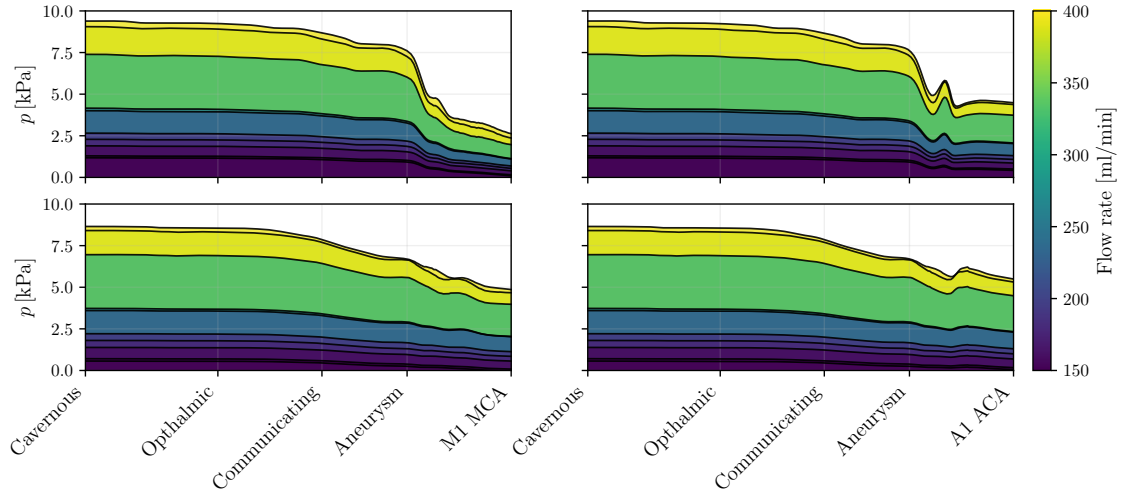
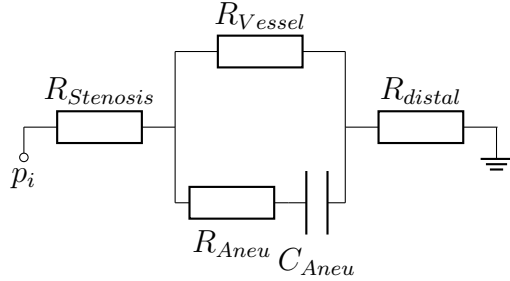


Figure 1.5: Pressures along the centerlines of the ICA and its daughter vessels ACA and MCA in the pre-stented (top) and post-stented (bottom) scenario. The shows the pressure prorgession in the centerline leading to the MCA whereas the right column does the same for the ACA. Note that here we only display the centerline pressure. It is fairly constant along the vessel cross-section excepting at the ICA terminus, where the flow split and the inertia can cause differences of up to 200 Pa on one slice.

1.4 Discussion

1.4.1 Circuit analogy



	Preop	Postop 1
$R_{Stenosis}$	high	0
R_{Vessel}	∞	low
R_{Aneu}	high	high
C_{Aneu}	0	high
R_{distal}	low	high

Figure 1.6: Schematic representation of the circuit with parallel and series resistances and capacitances, and corresponding dummy values for each component.

1.4.2 Literature cases

Summary Hirato[3]: - FDS assisted coiling - ICA giant 20-14mm - Rupture after 3 days (!) - No geometrical changes - No information about rupture site or thrombosis started - Two main reasons are cited, of which the first seems more likely in their case 1) Pressure increase due to removal of the stenosis and autoregulation 2) Wall necrosis due to

Summary Kulscar[6]: - analyzed 13 ruptured aneurysms (giant, saccular and fusiform) - Observed thrombus formation and subsequent rupture - patients had headaches before rupture - Patients in the late rupture group (n=3) had rupture before 5 months Conclusion: - Extensive thrombosis can lead to a sped up rupture and autolysis of the aneurysm walls - Doesn't necessarily apply here because of no thrombosis + our ruptures later

Summary Cebral[2]: - 3 giant aneurysms ruptured within days after stenting - 2 of them had stenosed sections at the aneurysm neck - Assumes unchanged inflow conditions, thus seeing a pressure increase after stenosis liberation - no PCOMs visible! - NEED TO CHECK KROU

Summary Rouchaud[7]: Literature review study showing results for aneurysm rupture and delayed intraparenchymal hemorrhages following FDS treatment. - Overall no new information, but probabilities. - All categories of our aneurysm fit well with the most common cases EXCEPT the delay time to the rupture, which is higher in our case.

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RÉSUMÉ

Sed tamen haec cum ita tutius observentur, quidam vigore artuum inminuto rogati ad nuptias ubi aurum dextris manibus cavatis offertur, inpigre vel usque Spoletium pergunt. haec nobilium sunt instituta.

MOTS CLÉS

Intracranial aneurysms, Haemodynamics, Computational Fluid Dynamics, Finite Elements, Variational Multi-scale Method

ABSTRACT

Ego vero sic intellego, Patres conscripti, nos hoc tempore in provinciis decernendis perpetuae pacis habere oportere rationem. Nam quis hoc non sentit omnia alia esse nobis vacua ab omni periculo atque etiam suspicione belli?

KEYWORDS

Intracranial aneurysms, Haemodynamics, Computational Fluid Dynamics, Finite Elements, Variational Multi-scale Method